BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. WHERE filters rows; ORDER BY arranges returned rows.

Q2. Sum 0, carry 1.

Q3. O(log n) for sorted random-access data.

Q4. On a match or after examining all items.

### You Do pages · Independent questions

Q1. INSERT INTO Student (StudentID, Name) VALUES (22, 'Kai'); [4].

Q2. UPDATE Student SET YearGroup=13 [3] WHERE StudentID=<specific existing ID> [2]; confirm selected row before applying [1].

Q3. Missing WHERE deletes all rows [2], FK/referential impacts [2], transactions/rollback and backups [2], permission controls and dry-run SELECT [2].

### Exit Ticket · Model answer

False. With no WHERE it updates every row.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

Do Now — Recall previous learning | I Do — Read new knowledge | We Do — Practise with guidance | You Do — Work independently | Review & Improve — Check understanding





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 139 · Lesson 139 · SQL INSERT, UPDATE and DELETE Safely

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the difference between WHERE and ORDER BY in a query?

Q2. What are the outputs of a half adder when A=1 and B=1?

Q3. In Big O notation, what is binary search worst-case complexity?

Q4. When does a linear search stop?

Do Now — Recall previous learning | I Do — Read new knowledge | We Do — Practise with guidance | You Do — Work independently | Review & Improve — Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

INSERT — Adds new rows.

UPDATE — Changes values in existing rows.

DELETE — Removes matching rows.

WHERE — Restricts affected rows; omission can affect every row.

### New learning

Create rows — INSERT INTO Table(columns) VALUES(...).

Change rows safely — UPDATE Table SET column=value WHERE key=...

Delete deliberately — DELETE FROM Table WHERE key=...; verify matches before writing.

### Worked example

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

1. Identify BookID 8 — Find the given data and name the required result.

2. UPDATE Stock = 3 — Explain why this intermediate stage is necessary.

3. WHERE BookID = 8 — UPDATE Book SET Stock = 3 WHERE BookID = 8; Before running, SELECT the matching row. Without WHERE, the stock column of all books would be changed.

## We Do · Practise with guidance

Add a new Club record and delete only the club with ClubID=12.

Insert club 24

Find the given data and name the required result.

Confirm club 12 exists

Explain why this intermediate stage is necessary.

DELETE WHERE ClubID=12

INSERT INTO Club (ClubID, Name) VALUES (24, 'Robotics'); DELETE FROM Club WHERE ClubID = 12; Foreign-key dependencies may prevent delete until related records are handled by defined policy.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Write SQL inserting StudentID 22 and Name 'Kai' into Student. [4]

### Question 2

A teacher must change a single pupil's YearGroup to 13. Explain a safe SQL statement. [6]

### Question 3

Explain risks in DELETE FROM Booking; and suggest protections in a relational production database. [8]

## Exit Ticket · Answer before checking

UPDATE Book SET Stock=0; changes exactly one Book record.